Part 6 Investment in Capital Assets

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What is Capital Budgeting?

The process of identifying, analyzing, and selecting investment projects whose returns (cash flows) are expected to extend beyond one year.

6.1.1. The Capital Budgeting Process

- Generate investment proposals consistent with the firm's strategic objectives.
- Estimate after-tax incremental operating cash flows for the investment projects.
- Evaluate project incremental cash flows.
- Select projects based on a value-maximizing acceptance criterion.
- Reevaluate implemented investment projects continually and perform postaudits for completed projects.

Classification of Investment Project Proposals

(1) New products or expansion of existing products

(2) Replacement of existing equipment or buildings

(3) Research and development

(4) Exploration

(5) Other (e.g., safety or pollution related)

Estimating After-Tax Incremental Cash Flows

- Basic characteristics of relevant project flows:
  - Cash (not accounting income) flows
  - Operating (not financing) flows
  - After-tax flows
  - Incremental flows
- Principles that must be adhered to in the estimation:
  - *Ignore* sunk costs
  - *Include* opportunity costs
  - *Include* project-driven changes in working capital net of spontaneous changes in current liabilities
  - *Include* effects of inflation

Tax Considerations and Depreciation

- *Depreciation* represents the systematic allocation of the cost of a capital asset over a period of time for financial reporting purposes, tax purposes, or both.
- Generally, profitable firms prefer to use an accelerated method for tax reporting purposes. Everything else equal, the greater the depreciation charges, the lower the taxes paid by the firm.
- Depreciation is a noncash expense.

Depreciable Basis

Depreciable Basis =

Cost of Asset + Capitalized Expenditures

- Capitalized Expenditures are expenditures that may provide benefits into the future and therefore are treated as capital outlays and not as expenses of the period in which they were incurred.
- *Examples*: Shipping and installation

Depreciation and the MACRS Method

- Everything else equal, the greater the depreciation charges, the lower the taxes paid by the firm.
- Depreciation is a noncash expense.
- Assets are depreciated (MACRS) on one of eight different property classes.
- Generally, the half-year convention is used for MACRS.

MACRS Sample Schedule

Recovery Year	Property Class		
	3-Year	5-Year	7-Year
1	33.33%	20.00%	14.29%
2	44.45	32.00	24.49
3	14.81	19.20	17.49
4	7.41	11.52	12.49
5		11.52	8.93
6		5.76	8.92
7			8.93
8			4.46

Calculating the Incremental Cash Flows

- Initial cash outflow – the initial net cash investment.
- Interim incremental net cash flows – those net cash flows occurring after the initial cash investment but not including the final period's cash flow.
- Terminal-year incremental net cash flows – the final period's net cash flow.

Initial Cash Outflow

a) *Cost of "new" assets*

b) + Capitalized expenditures

c) + (–) Increased (decreased) NWC

d) – Net proceeds from sale of "old" asset(s) if replacement

e) + (–) Taxes (savings) due to the sale of "old" asset(s) if replacement

f) = Initial cash *outflow*

Incremental Cash Flows

a) Net incr. (decr.) in operating revenue less (plus) any net incr. (decr.) in operating expenses, excluding depr.
b) – (+) Net incr. (decr.) in tax depreciation
c) = Net change in income before taxes
d) – (+) Net incr. (decr.) in taxes
e) = Net change in income after taxes
f) + (–) Net incr. (decr.) in tax depr. charges
g) = Incremental net cash flow for period

Terminal-Year Incremental Cash Flows

a) Calculate the incremental net cash flow for the terminal period

b) + (–) Salvage value (disposal/reclamation costs) of any sold or disposed assets

c) – (+) Taxes (tax savings) due to asset sale or disposal of "new" assets

d) + (–) Decreased (increased) level of "net" working capital

e) = Terminal year incremental net cash flow

Example of an Asset Expansion Project

- The machine will cost $60,000 plus $20,000 for shipping and installation and depreciated using straight-line over 4 years . NWC will rise by $5,000. Revenues will increase by $110,000 for each of the next 4 years and will then be sold (scrapped) for $10,000 at the end of the fourth year, when the project ends. Operating costs will rise by $70,000 for each of the next four years. Income tax 40%.

Initial Cash Outflow (ICO)

a)	+	Cost of “new” assets	$60,000
b)	+	Capitalized expenditures	$20,000
c)	+	Increased (decreased) NWC	$5,000
d)	-	Net proceeds from sale of “old” asset(s)	$0
e)	+	Taxes due to the sale of “old” asset(s)	$0
f)		Initial Cash Outflow (ICO)	$85,000

Incremental Cash Flows

		Year 1	Year 2	Year 3	Year 4
a)		$40,000	$40,000	$40,000	$40,000
b)	–	20,000	20,000	20,000	20,000
c)	=	$20,000	$20,000	$20,000	$20,000
d)	–	8,000	8,000	8,000	8,000
e)	=	$12,000	$12,000	$12,000	$12,000
f)	+	20,000	20,000	20,000	20,000
g)	=	$32,000	$32,000	$32,000	$32,000

Terminal-Year Incremental Cash Flows

a) $32,000 The incremental cash flow from the previous slide in Year 4.

b) + 10,000 Salvage Value.

c) – 4,000 .40*($10,000 - 0) Note, the asset is fully depreciated at the end of Year 4.

d) + 5,000 NWC - Project ends.

e) = $43,000 Terminal-year incremental cash flow.

Example of an Asset Replacement Project

- The original basis of the machine was $30,000 and depreciated using straight-line. The machine has three years of depreciation and five years of useful life remaining. BW can sell the current machine for $6,000. The new machine will cost $60,000 plus $20,000 for shipping and installation, and depreciated using straight-line over 4 years. The new machine will not increase revenues (remain at $110,000) but it decreases operating expenses by $10,000 per year (old = $80,000). NWC will rise to $10,000 from $5,000 (old). Salvage value is 10,000. Income tax 40%.

Initial Cash Outflow

a)		$60,000	
b)	+	20,000	
c)	+	5,000	(=10,000 – 5,000)
d)	–	6,000	(sale of “old” asset)
e)	+	2,400	<---- Tax 40%
f)	=	$81,400	

Calculation of the Change in Depreciation

		Year 1	Year 2	Year 3	Year 4
a)		$20,000	$20,000	$20,000	$20,000
b)	–	6,000	6,000	0	0
c)	=	$14,000	$14,000	$20,000	$20,000

a) Represent the depreciation on the "new" project.

b) Represent the remaining depreciation on the "old" project.

c) Net *change* in tax depreciation charges.

Incremental Cash Flows

		Year 1	Year 2	Year 3	Year 4
a)		$10,000	$10,000	$10,000	$10,000
b)	–	14,000	14,000	20,000	20,000
c)	=	$ –4,000	$ -4,000	$-10,000	$-10,000
d)	–	–1,600	–1,600	-4,000	-4,000
e)	=	$ –3,400	$ –3,400	$ -6,000	$ -6,000
f)	+	14,000	14,000	20,000	20,000
g)	=	$11,600	$11,600	$14,000	$ 14,000

Terminal-Year Incremental Cash Flows

a) $ 14,000 The incremental cash flow from the previous slide in Year 4.

b) + 10,000 Salvage Value.

c) – 4,000 (.40)*($10,000 – 0). Note, the asset is fully depreciated at the end of Year 4.

d) + 5,000 Return of "added" NWC.

e) = $25,000 Terminal-year incremental cash flow.

Summary of Project Net Cash Flows

Asset Expansion

Year 0   Year 1   Year 2   Year 3   Year 4

–$85,000* $32,000 $32,000 $32,000
$43,000

Asset Replacement

Year 0   Year 1   Year 2   Year 3   Year 4

–$76,600* $11,600 $11,600 $14,000
$25,000

The Faversham Fish Farm is considering the introduction of a new fish-flaking facility. To launch the facility, it will need to spend $90,000 for special equipment. The equipment has a useful life of four years and depreciated using straight-line over 4 years. Shipping and installation expenditures equal $10,000, and the machinery has an expected final salvage value, four years from now, of $16,500. The marketing department envisions that use of the new facility will generate additional net operating revenue cash flows, before consideration of depreciation and taxes will be $35,167, $36,250, $55,725 and $32 258, respectively, for each of the Years 1 through 4.

Assuming that the marginal tax rate equals 40 percent, we now need to estimate the project's relevant incremental cash flows.

The project's initial cash outflow (ICO):

Cost of equipment	$90,000
+ Shipping and installation expenditures	$10,000
= ICO	**$100,000**

The incremental net cash flows:

	1	2	3	4
Net change in operating evalue	$35,167	$36,250	$55,725	$32,258
- Net increase in tax depreciation charges	$25,000	$25,000	$25,000	$25,000
= Net change in income before taxes	$10,167	$11,250	$30,725	$7,258
-(+) Net inc. (dec.) in taxes (40% rate)	$4,066.8	$4,500	$8,290	$2,903.2
= Net change in income after taxes	$6,100.2	$6,750	$12,435	$4,354.8
+ Net increase in tax depreciation charges	$25,000	$25,000	$25,000	$25,000
= Incremental net cash flows	**$31,100.2**	**$31,750**	**$37,435**	**$29,354.8**

Terminal-year incremental net cash flow:

Incremental cash flow for the terminal year before project windup considerations	$29,354.8
+ Final salvage value of the machinery	$16,500
- Taxes due to sale or disposal	$6,600
Terminal-year incremental net cash flow	**$39,254.8**

The expected incremental net cash flows from the project are

	1	2	3	4
Net cash flows	$31,100.2	$31,750	$37,435	$39,254.8

6.2. Project Evaluation and Selection

Alternative Methods

6.2.1 Payback Period (PBP)

6.2.2 Internal Rate of Return (IRR)

6.2.3 Net Present Value (NPV)

6.2.4 Profitability Index (PI)

Proposed Project Data

- Julie Miller is evaluating a new project for her firm, *Basket Wonders (BW)*.
- She has determined that the after-tax cash flows for the project will be $10,000; $12,000; $15,000; $10,000; and $7,000, respectively, for each of the Years 1 through 5. The initial cash outlay will be $40,000.

Independent Project

- For this project, assume that it is *independent* of any other potential projects that *Basket Wonders* may undertake.
- *Independent* – A project whose acceptance (or rejection) does not prevent the acceptance of other projects under consideration.

Payback Period (PBP)

0	1	2	3	4	5
-40 K	10 K	12 K	15 K	10 K	7 K

PBP is the period of time required for the cumulative expected cash flows from an investment project to equal the initial cash outflow.

Payback Period (PBP)

0	1	2	3 (a)	4	5
-40 K (-b)	10 K	12 K	15 K	*10 K* (d)	7 K
	10 K	22 K	*37 K* (c)	47 K	54 K

Cumulative Inflows

PBP = a + (b - c) / d =

3 + (40 - 37) / 10 = 3 +

(3) / 10 = 3.3 Years

Payback Period (PBP)

0	1	2	3	4	5
-40 K	10 K	12 K	15 K	10 K	7 K
-40 K	-30 K	-18 K	-3 K	7 K	14 K

Cumulative Cash Flows

PBP = 3 + (3K) / 10K

= 3.3 Years

Note: Take absolute value of last negative cumulative cash flow value.

PBP Acceptance Criterion

The management of *Basket Wonders* has set a maximum PBP of 3.5 years for projects of this type.

Should this project be accepted?

Yes!

- The firm will receive back the initial cash outlay in less than 3.5 years. [3.3 Years < 3.5 Year Max.]

FMS

PBP Strengths and Weaknesses

Strengths:

- Easy to use and understand
- Can be used as a measure of liquidity
- Easier to forecast Short-Term than Long-Term flows

Weaknesses:

- Does not account for Time value of money
- Does not consider cash flows beyond the PBP
- Cutoff period is subjective

Internal Rate of Return (IRR)

- IRR is the discount rate that equates the present value of the future net cash flows from an investment project with the project's initial cash outflow.

$$ICO = \frac{CF_1}{(1 + IRR)^1} + \frac{CF_2}{(1 + IRR)^2} + \ldots + \frac{CF_n}{(1 + IRR)^n}$$

IRR Solution

$$\$40{,}000 = \frac{\$10{,}000}{(1+IRR)^1} + \frac{\$12{,}000}{(1+IRR)^2} + \frac{\$15{,}000}{(1+IRR)^3} + \frac{\$10{,}000}{(1+IRR)^4} + \frac{\$7{,}000}{(1+IRR)^5}$$

Find the interest rate (*IRR*) that causes the discounted cash flows to equal $40,000.

[1] Try Ms.Excel: irr = 11.47%

[2] Try interpolation:

$$irr = i_1 + \frac{NPV_1 \times (i_2 - i_1)}{/NPV_1/ + /NPV_2/}$$

IRR Solution

$\$40,000 = \$10,000(PVIF_{10\%,1}) + \$12,000(PVIF_{10\%,2}) + \$15,000(PVIF_{10\%,3}) + \$10,000(PVIF_{10\%,4}) + \$7,000(PVIF_{10\%,5})$

$\$40,000 = \$10,000(.909) + \$12,000(.826) + \$15,000(.751) + \$10,000(.683) + \$7,000(.621)$

$\$40,000 = \$9,090 + \$9,912 + \$11,265 + \$6,830 + \$4,347 = \$41,444$ [*Rate is too low!!*]

FMS

IRR Solution

$\$40,000 = \$10,000(PVIF_{15\%,1}) + \$12,000(PVIF_{15\%,2}) + \$15,000(PVIF_{15\%,3}) + \$10,000(PVIF_{15\%,4}) + \$7,000(PVIF_{15\%,5})$

$\$40,000 = \$10,000(.870) + \$12,000(.756) + \$15,000(.658) + \$10,000(.572) + \$7,000(.497)$

$\$40,000 = \$8,700 + \$9,072 + \$9,870 + \$5,720 + \$3,479 = \$36,841$ [*Rate is too high!!*]

IRR Solution (Interpolate)

	.10	$41,444		
.05	IRR	$40,000	X	$1,444
	.15	$36,841		$4,603

$$\frac{X}{.05} = \frac{\$1,444}{\$4,603}$$

IRR Solution (Interpolate)

.10 $41,444

.05 X IRR $40,000 $1,444 $4,603

.15 $36,841

$$\frac{X}{.05} = \frac{\$1,444}{\$4,603}$$

IRR Solution (Interpolate)

.10 $41,444

.05 IRR $40,000 X $1,444 $4,603

.15 $36,841

$$X = \frac{(\$1{,}444)(0.05)}{\$4{,}603} \qquad X = .0157$$

IRR = .10 + .0157 = .1157 or 11.57%

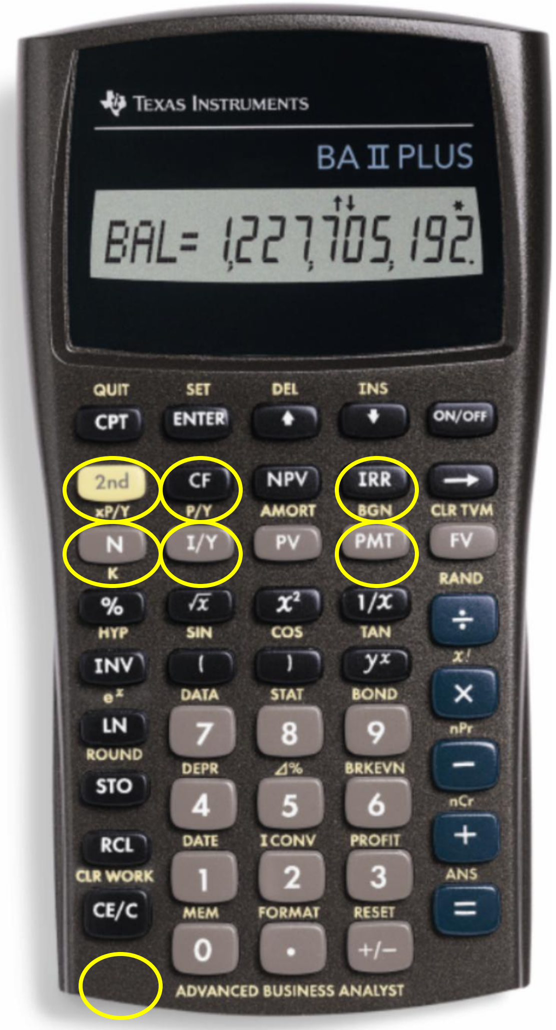
IRR Acceptance Criterion

The management of *Basket Wonders* has determined that the hurdle rate is 13% for projects of this type.

Should this project be accepted?

No! The firm will receive 11.47% for each dollar invested in this project at a cost of 13%. [IRR < Hurdle Rate]

IRRs on the Calculator



We will use the cash flow registry to solve the IRR for this problem quickly and accurately!

Actual IRR Solution Using Your Financial Calculator

Steps in the Process

Step 1:	Press	CF		key		
Step 2:	Press	2^{nd} CLR Work		keys		
Step 3:	For *CF0* Press	-40000		Enter	↓	keys
Step 4:	For *C01* Press	10000		Enter	↓	keys
Step 5:	For *F01* Press	1		Enter	↓	keys
Step 6:	For *C02* Press	12000		Enter	↓	keys
Step 7:	For *F02* Press	1		Enter	↓	keys
Step 8:	For *C03* Press	15000		Enter	↓	keys
Step 9:	For *F03* Press	1		Enter	↓	keys

Actual IRR Solution Using Your Financial Calculator

Steps in the Process (Part II)

Step 10:	For *C04*	Press	10000	Enter	↓	keys
Step 11:	For *F04*	Press	1	Enter	↓	keys
Step 12:	For *C05*	Press	7000	Enter	↓	keys
Step 13:	For *F05*	Press	1	Enter	↓	keys
Step 14:		Press	↓ ↓			keys
Step 15:		Press	IRR			key
Step 16:		Press	CPT			key

Result: Internal Rate of Return = 11.47%

IRR Strengths and Weaknesses

Strengths:

- Accounts for TVM
- Considers all cash flows
- Less subjectivity

Weaknesses:

- Assumes all cash flows reinvested at the IRR
- Difficulties with project rankings and Multiple IRRs

Net Present Value (NPV)

NPV is the present value of an investment project's net cash flows minus the project's initial cash outflow.

$$NPV = \frac{CF_1}{(1+k)^1} + \frac{CF_2}{(1+k)^2} + \cdots + \frac{CF_n}{(1+k)^n} - ICO$$

NPV Solution

Basket Wonders has determined that the appropriate discount rate (k) for this project is 13%.

$$NPV = \frac{\$10,000}{(1.13)^1} + \frac{\$12,000}{(1.13)^2} + \frac{\$15,000}{(1.13)^3} +$$

$$\frac{\$10,000}{(1.13)^4} + \frac{\$7,000}{(1.13)^5} - \$40,000$$

NPV Solution

$$NPV = \$10,000(PVIF_{13\%,1}) + \$12,000(PVIF_{13\%,2}) + \$15,000(PVIF_{13\%,3}) + \$10,000(PVIF_{13\%,4}) + \$7,000(PVIF_{13\%,5}) - \$40,000$$

$$NPV = \$10,000(.885) + \$12,000(.783) + \$15,000(.693) + \$10,000(.613) + \$7,000(.543) - \$40,000$$

$$NPV = \$8,850 + \$9,396 + \$10,395 + \$6,130 + \$3,801 - \$40,000$$

$$= -\$1,424$$

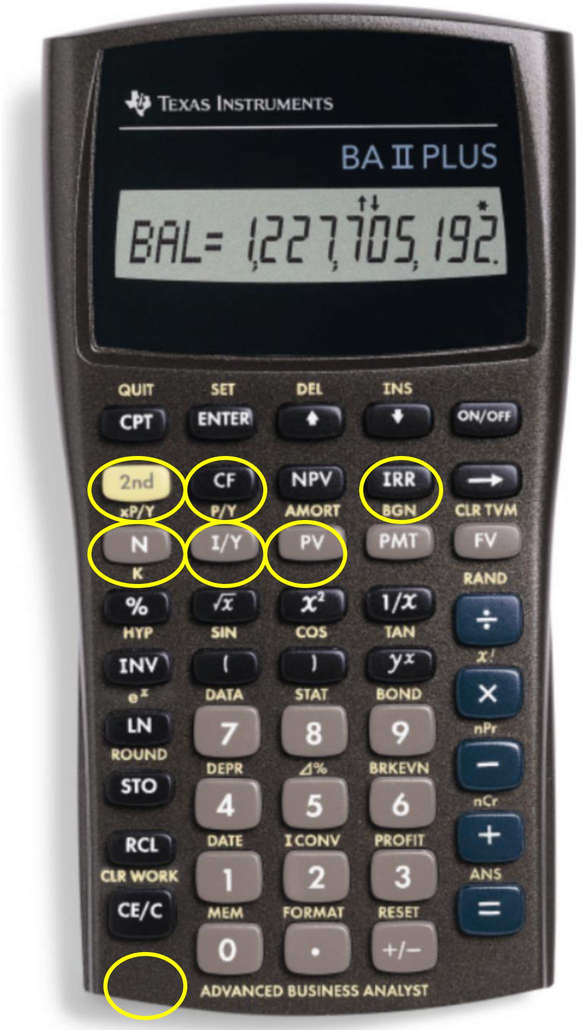
NPV Acceptance Criterion

The management of *Basket Wonders* has determined that the required rate is 13% for projects of this type.

Should this project be accepted?

No! The NPV is *negative*. This means that the project is reducing shareholder wealth. [*Reject* as *NPV* < *0*]

NPV on the Calculator



We will use the cash flow registry to solve the NPV for this problem quickly and accurately!

<u>Hint</u>: If you have *<u>not</u>* cleared the cash flows from your calculator, then you may skip to Step 15.

Actual NPV Solution Using Your Financial Calculator

Steps in the Process

Step 1:	Press	CF		key		
Step 2:	Press	2^{nd} CLR Work		keys		
Step 3:	*For CF0* Press	-40000	Enter	↓	keys	
Step 4:	*For C01* Press	10000	Enter	↓	keys	
Step 5:	*For F01* Press	1	Enter	↓	keys	
Step 6:	*For C02* Press	12000	Enter	↓	keys	
Step 7:	*For F02* Press	1	Enter	↓	keys	
Step 8:	*For C03* Press	15000	Enter	↓	keys	
Step 9:	*For F03* Press	1	Enter	↓	keys	

Actual NPV Solution Using Your Financial Calculator

Steps in the Process (Part II)

Step 10: For *C04* Press 10000 Enter ↓ keys
Step 11: For *F04* Press 1 Enter ↓ keys
Step 12: For *C05* Press 7000 Enter ↓ keys
Step 13: For *F05* Press 1 Enter ↓ keys

Step 14: Press ↓ ↓ keys

Step 15: Press NPV key

Step 16: For I=, Enter 13 Enter ↓ keys

Step 17: Press CPT key

Result: Net Present Value = -$1,424.42

NPV Strengths and Weaknesses

Strengths:

- Cash flows assumed to be reinvested at the hurdle rate.
- Accounts for TVM.
- Considers all cash flows.

Weaknesses:

- Not easy to apply

Profitability Index (PI)

PI is the ratio of the present value of a project's future net cash flows to the project's initial cash outflow.

$$PI = \left[\frac{CF_1}{(1+k)^1} + \frac{CF_2}{(1+k)^2} + \ldots + \frac{CF_n}{(1+k)^n} \right] \div ICO$$

PI Acceptance Criterion

PI = $38,572 / $40,000

= .9643

Should this project be accepted?

No! The PI is *less than 1.00*. This means that the project is not profitable. [*Reject* as *PI* < *1.00*]

PI Strengths and Weaknesses

Strengths:

- Same as NPV
- Allows comparison of different scale projects

Weaknesses:

- Same as NPV
- Provides only relative profitability
- Potential Ranking Problems

Evaluation Summary

Basket Wonders Independent Project

Method	Project	Comparison	Decision
PBP	3.3	3.5	Accept
IRR	11.47%	13%	Reject
NPV	-$1,424	$0	Reject
PI	.96	1.00	Reject

6.3. Other Project Relationships

- *Dependent* – A project whose acceptance depends on the acceptance of one or more other projects.
- *Mutually Exclusive* -- A project whose acceptance precludes the acceptance of one or more alternative projects.
- Ranking of project proposals *may* create contradictory results:
  A. Scale of Investment
  B. Cash-flow Pattern
  C. Project Life

A. Scale Differences

Compare a small (S) and a large (L) project.

	NET CASH FLOWS	
END OF YEAR	Project S	Project L
0	-$100	-$100,000
1	0	
0		
2	$400	$156,250

A. Scale Differences

Calculate the PBP, IRR, NPV@10%, and PI@10%.

Which project is preferred? Why?

Project	IRR	NPV	PI
S	100%	$ 231	3.31
L	25%	$29,132	1.29

B. Cash Flow Pattern

Let us compare a *decreasing* cash-flow (D) project and an *increasing* cash-flow (I) project.

		NET CASH FLOWS	
	END OF YEAR	Project D	Project I
	0	-$1,200	-$1,200
	1	1,000	
100	2	500	
600	3	100	
1,080			

Cash Flow Pattern

Calculate the IRR, NPV@10%, and PI@10%.

Which project is preferred?

Project	IRR	NPV	PI
D	23%	$198	1.17
I	17%	$198	1.17

C. Project Life Differences

Let us compare a *long* life (X) project and a *short* life (Y) project.

	END OF YEAR	NET CASH FLOWS Project X	NET CASH FLOWS Project Y
	0	-$1,000	-$1,000
	1	0	
2,000	2	0	
0	3	3,375	
0			

Project Life Differences

Calculate the PBP, IRR, NPV@10%, and PI@10%.

Which project is preferred? Why?

Project	IRR	NPV	PI
X	50%	$1,536	2.54
Y	100%	$ 818	1.82

Another Way to Look at Things

1. Adjust cash flows to a common terminal year if project "Y" will *NOT* be replaced.

Compound Project Y, Year 1 @10% for 2 years.

Year	0	1	2	3
CF	-$1,000	$0	$0	$2,420

Results: IRR* = 34.26%   NPV = $818

**Lower IRR* from adjusted cash-flow stream. X is still Best.

Replacing Projects with Identical Projects

2. Use *Replacement Chain Approach* (Appendix B) when project "Y" will be replaced.

0	1	2	3
-$1,000	$2,000		
	-1,000	$2,000	
		-1,000	$2,000
-$1,000	$1,000	$1,000	$2,000

Results: IRR* = 100% NPV* = $2,238.17

**Higher NPV, but the same IRR*. Y is Best.

Capital Rationing

Capital Rationing occurs when a constraint (or budget ceiling) is placed on the total size of capital expenditures during a particular period.

Example: Julie Miller must determine what investment opportunities to undertake for *Basket Wonders (BW)*. She is limited to a maximum expenditure of $32,500 *only* for this capital budgeting period.

Available Projects for BW

Project	ICO	IRR	NPV	PI
A	$ 500	18%	$ 50	1.10
B	5,000	25	6,500	2.30
C	5,000	37	5,500	2.10
D	7,500	20	5,000	1.67
E	12,500	26	500	1.04
F	15,000	28	21,000	2.40
G	17,500	19	7,500	1.43
H	25,000	15	6,000	1.24

Choosing by IRRs for BW

Project ICO IRR NPV PI

C $ 5,000 37% $5,500 2.10 F
15,000 28 21,000 2.40 E 12,500 26
500 1.04 B 5,000 25 6,500 2.30

Projects C, F, and E have the three *largest IRRs*.

The resulting *increase* in shareholder wealth is $27,000 with a $32,500 outlay.

Choosing by NPVs for BW

Project ICO IRR NPV PI

F $15,000 28% $21,000 2.40 G

17,500 19 7,500 1.43 B 5,000 25

6,500 2.30

Projects F and G have the two *largest NPVs*.

The resulting *increase* in shareholder wealth is $28,500 with a $32,500 outlay.

Choosing by PIs for BW

Project	ICO	IRR	NPV	PI
F	$15,000	28%	$21,000	2.40
B	5,000	25	6,500	2.30
C	5,000	37	5,500	2.10
D	7,500	20	5,000	1.67
G	17,500	19	7,500	1.43

Projects F, B, C, and D have the four *largest PIs*.

The resulting *increase* in shareholder wealth is $38,000 with a $32,500 outlay.

Summary of Comparison

Method	Projects Accepted	Value Added
PI	F, B, C, and D	$38,000
NPV	F and G	$28,500
IRR	C, F, and E	$27,000

PI generates the *greatest increase* in *shareholder wealth* when a limited capital budget exists for a *single period*.

Post-Completion Audit

Post-completion Audit

A formal comparison of the actual costs and benefits of a project with original estimates.

- Identify any project weaknesses
- Develop a possible set of corrective actions
- Provide appropriate feedback

Result: Making better future decisions!

Multiple IRR Problem*

Let us assume the following cash flow pattern for a project for Years 0 to 4:

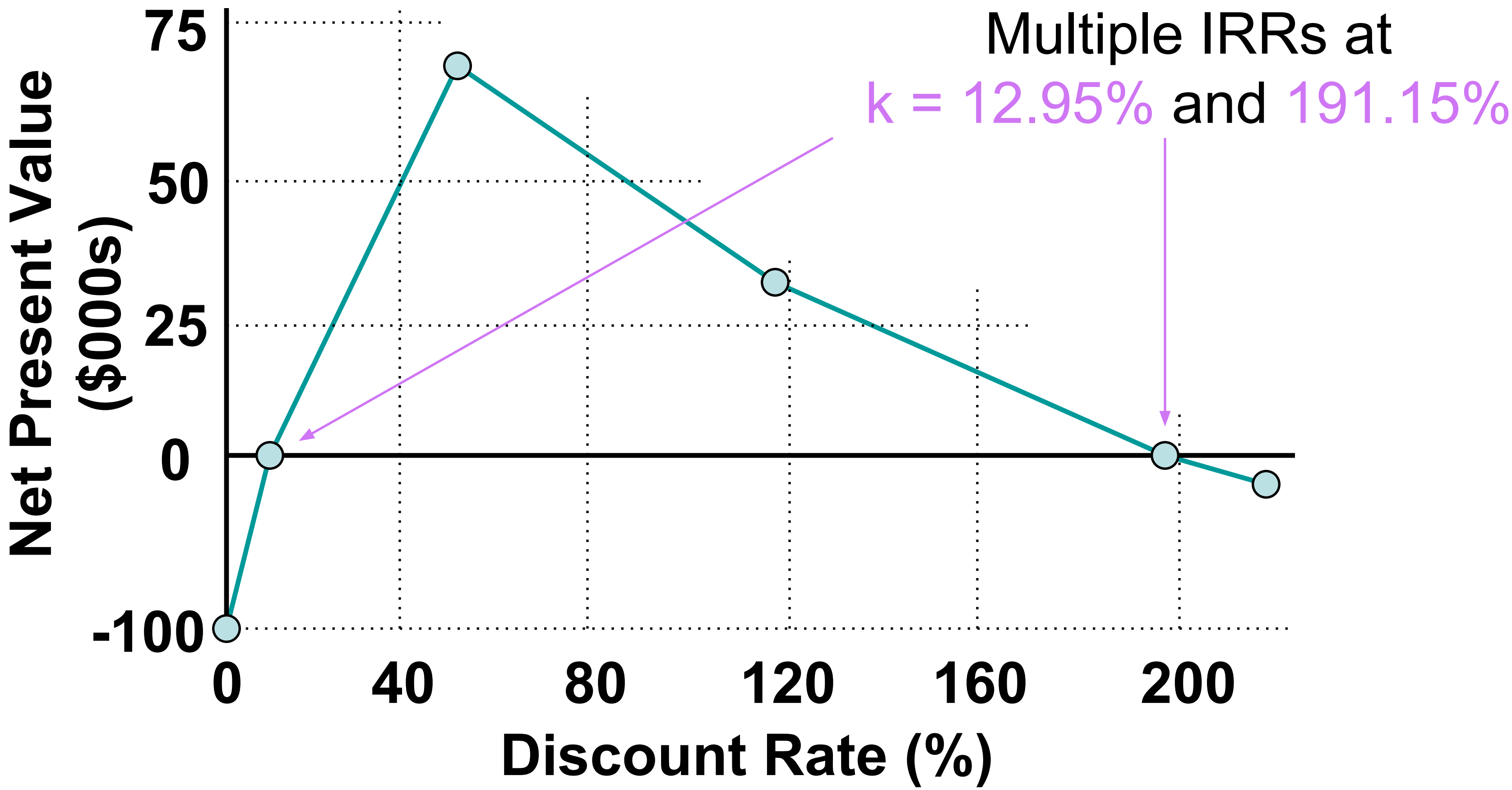
-$100 +$100 +$900 -$1,000

How many *potential* IRRs could this project have?

Two!! There are as many potential IRRs as there are sign changes.

**** Refer to Appendix A***

NPV Profile -- Multiple IRRs



Company is considering the replacement of two machines that are two years old with a new, more efficient machine. The two old machines could be sold currently for a total of $70,000 in the secondary market, but they would have a zero final salvage value if held to the end of their remaining useful life. Their original depreciable basis totaled $300,000. They have a depreciated tax book value of $86,400, and a remaining useful life of eight years. Straight-line depreciation is used on these machines. The new machine can be purchased and installed for $480,000. It has a useful life of eight years, at the end of which a salvage value of $40,000 is expected. Owing to its greater efficiency, the new machine is expected to result in incremental annual operating savings of $100,000. The company's corporate tax rate is 40 percent, and if a loss occurs in any year on the project, it is assumed that the company can offset the loss against other company income.

- What are the incremental cash inflows over the eight years, and what is the incremental cash outflow at time 0?
- If opportunity cost of funds is 10%, calculate NPV, IRR and PI? Should Company accept this project?

- X has determined that the earning before-tax for the project will be: $51,500; $53,000; $53,000; $49,000, respectively, for each of the Years 1 through 4. The initial cash outlay will be $500,000. Final salvage value of asset is $10,000. The Company expects to depreciate its assets on a straight-line basis. Assuming that the marginal tax rate equals 20% (not on salvage value of asset). What are the incremental cash inflows over the four years, and what is the incremental cash outflow at time 0?
- If opportunity cost of funds is 15%, calculate NPV, IRR and PI? Should X accept this project?

1. A project with the initial investment of VND 450 million has an operating period of 4 years. The earning after-tax through 4 years is VND 120; 180; 190; 160 million respectively. Calculate the IRR of this project.

2. Company X has determined that the earning after tax for the project of buying a new machine will be VND 56.25; 60; 60; 52.5 million respectively, for each of the Years 1 through 4. The initial investment is VND 500 million. Final salvage value of the machine after 4 year is VND 12.5 million. Assuming that the marginal tax rate equals 25% (not on salvage value of asset). The Company expects to depreciate its assets on a straight-line basis. The cost of capital of the company is 15% per year. What is the net present value (NPV) of this project?

3. X has determined that the before-tax cash flows for the project will be \$35,000; \$30,000; \$40,000; \$38,000; \$30,000, respectively, for each of the Years 1 through 5. The initial cash outlay will be \$350,000. Final salvage value of asset is \$15,000. Assuming that the marginal tax rate equals 28% (not on salvage value of asset). The Company expects to depreciate its assets on a straight-line basis. What are the incremental cash inflows over the 5 years, and what is the incremental cash outflow at time 0? Calculate NPV, IRR and PI? Should X accept this project? Assuming the cost of capital of the company is 15% per year.

4. X has determined that the earning before-tax for the project will be: \$35,000; \$30,000; \$40,000; \$38,000, respectively, for each of the Years 1 through 4. The initial cash outlay will be \$350,000. Final salvage value of asset is zero. The Company expects to depreciate its assets on a straight-line basis. Assuming that the marginal tax rate equals 28%. What is the Net Present Value (NPV) at 10% discount rate?